

Geographical Coarse Graining of Complex Networks

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We perform a renormalization-grouplike numerical analysis of geographically embedded complex networks on a two-dimensional square lattice. At each step of the coarse-graining procedure, the four vertices on each 2×2 square box are merged to a single vertex, resulting in a coarse-grained system of smaller size. Repetition of the process leads to the observation that the coarse-graining procedure does not alter the qualitative characteristics of the original scale-free network, which opens the possibility of subtracting a smaller network from the original network without destroying the important structural properties. The implication of the result is also suggested in the context of the recent study of the human brain functional network.

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The study of complex networks has been one of the most active research areas not only in physics but also in other various disciplines of natural and social sciences [1–3]. In some existing networks, the computerized automatic data acquisition techniques make it possible to grab the detailed information of interconnections in networks. In contrast, in many biological and social networks, the complete network structure is hard to define and even when it is possible it requires tremendous time-demanding efforts. The detailed structure of the neuronal network of *Caenorhabditis elegans* [2], composed of about 300 neuron cells and 14 synaptic couplings per neuron, has been obtained by biologists through direct observations. In comparison to the *C. elegans* neural network, the complexity of the human brain is gigantic: It contains about 10^{11} neuron cells, each of which is connected to 10^3 – 10^4 other neurons via synaptic couplings. The construction of the detailed map of all neuron connections in the human brain is beyond imagination and will be so in the future.

In the viewpoint of statistical physics, on the other hand, understanding the qualitative collective behavior of the brain, although it originates from the actual detailed interactions of neurons and abundant biochemical substances, may not require such a detailed microscopic map of interneuron connections. In this regard, the recent study by Eguíluz *et al.* in Ref. [4] draws much interest: Brain activity has been measured from $32 \times 64 \times 64$ sites (called voxels) and the intervoxel correlation has been used to map out the functional network of the human brain. Although the number of voxels in Ref. [4] is more than a million, each voxel still contains $O(10^5)$ neuron cells. Consequently, one can say that the brain functional network in Ref. [4] has been based on *heavily coarse-grained* information, and thus it is not clear whether the observed scale freeness of the network is a genuine emerging property of actual interneuron connections or not. Very recently it has been found that a very simple vertex merging process results in scale-free networks,

regardless of the initial network structure [5]. In the present context, this observation may suggest that if the network is too coarse-grained, one cannot trust the resulting scale-free distribution since it may not reflect the structure of the original network but is a simple artifact of coarse graining.

In this Letter, we start from model scale-free networks that are geographically embedded [6], and then repeat several steps of geographic coarse graining. We find that the coarse-graining process does not change important properties of the original network. In particular, the degree exponent γ , the clustering property, the assortative feature, and the hierarchical structure do not change much upon the iteration of the geographic coarse graining. Our result suggests that the scale-free feature of the human brain functional network may not be the artifact of coarse graining, and thus the increase (or the decrease) of the size of voxels is expected not to change the main results of Ref. [4]. We also suggest that one can use the geographic coarse-graining method presented in this Letter to subtract a smaller network from the original larger network, without destroying important structural properties. This can be very useful when the network is too big to be handled for a given computational capability.

We first build the geographically embedded scale-free network following Ref. [6]: $N = L \times L$ vertices are put on lattice points of the two-dimensional square net, and then the degree k of each vertex is chosen according to the degree distribution function $p(k) \propto k^{-\gamma}$. A vertex v is selected at random and then its assigned degree k_v is realized on the basis that the geographically closer vertices (within the distance proportional to $\sqrt{k_v}$) are connected first. As the procedure is repeated over vertices, some vertices may not fulfill their assigned degrees if all their possible target vertices have already exhausted their allowed number of edges. When this happens, those vertices have degrees different from the initially assigned ones, and there appears the cutoff degree scale (and the corresponding cutoff length scale) beyond which $p(k)$

deviates from the power-law form $p(k) \sim k^{-\gamma}$ (see Ref. [6] for details).

Once the network is constructed in this way, we repeat the following geographic coarse graining, which is in parallel to the Kadanoff block spin renormalization-group procedure in standard statistical mechanical systems (see Fig. 1): Four vertices on each square box of the size 2×2 are merged to a single vertex and accordingly the edges connecting intrabox vertices (dashed lines in Fig. 1) are disregarded, but the interbox connections are kept (thin solid line in Fig. 1). We also keep track of how strong the edges are by assigning the weight w_{vw} that is simply the number of edges connecting two merged vertices v and w . For example, in Fig. 1 there exist two edges (thin solid line) connecting the two square boxes before the merging, which gives rise to the weight $w = 2$ for the edge (thick solid line) connecting the two vertices after the merging.

If we keep all the edges in the coarse-grained network, the average degree increases as the procedure is iterated, resulting in the fully connected network eventually. To remedy this, we fix the average degree at each step of coarse graining by removing weaker edges with smaller values of the weight. Suppose that we have to remove M_r edges to keep the average degree the same, and that there are M_w edges of the weight w . For example, for $M_r < M_{w=1}$, randomly picked M_r edges of $w = 1$ are removed. If $M_{w=1} < M_r < M_{w=2}$, all $w = 1$ edges are removed and $M_r - M_{w=1}$ edges with $w = 2$ are randomly deleted. The above procedure makes sense since in real situations it is common that the coarse graining is often accompanied by the change of the sensitivity of the measurement: When the system is looked at from a far distance, we only have interest in large scale structures.

Figure 2 shows the result of the coarse graining. Original networks of the size 64×64 are generated following Ref. [6] for $\gamma = 2.5, 3.0, 4.0, 5.0$ and the above explained coarse-graining process is iterated. The net-

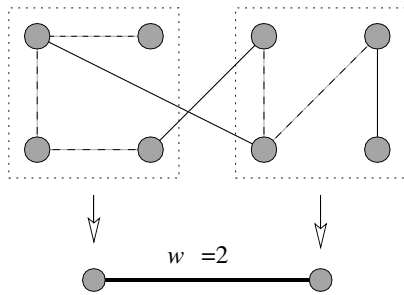


FIG. 1. Coarse-graining procedure. Four vertices in each 2×2 box are merged to a single vertex. After merging, two edges (thin solid lines) connecting vertices in different boxes become one edge (thick solid line) with the weight $w = 2$. This procedure is repeated for all 2×2 square boxes resulting in the coarse-grained network which is 4 times smaller than the original network. We then remove edges of smaller weights in order to fix the average degree.

work size in this work is smaller than the cutoff length scale beyond which the network ceased to be scale-free, which is also seen in Fig. 2(a) where the cutoff degree scale is absent (see Ref. [6]). One sees clearly that the coarse-graining process does not change the degree exponent γ . In the terminology of the renormalization-group (RG) formalism, the scale-free network with any value of γ is the stable fixed point of the RG flow. This observation implies that the scale-free network in Ref. [6] possesses neither the degree scales nor the length scales up to the cutoff length scale which is larger than the network size in the present study. In Fig. 2, as the coarse-graining process is iterated the scale-free region appears only for sufficiently large k regions.

We then study finite-size effects in Fig. 3, which shows $p(k)$ at the n th iterations for initial networks of various sizes (a) 32×32 , (b) 64×64 , (c) 96×96 , and (d) 128×128 (all for $\gamma = 3$). Clearly exhibited is that as the initial network size is increased the network still remains scale-free even after many steps of iterations, which then excludes the possibility that observed behaviors are finite-size artifacts. The scale-free degree distribution detected in the human brain *functional* network [4] does not actually imply that the neural network of human brain is scale-free. One reason is because the technique in Ref. [4] only measures the functionality correlation of two separate voxels, not the actual path of voxels through which biochemical signal transfers. One can also argue that since each voxel contains a large number of neurons [about $O(10^5)$], the observed scale-free distributions can be the artifact of the coarse grain-

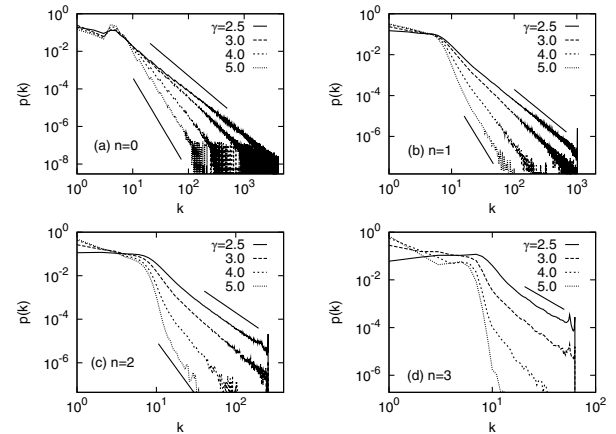


FIG. 2. Degree distribution $p(k)$ versus the degree k for geographically embedded scale-free networks. The original 64×64 networks in (a) with the degree exponents $\gamma = 2.5, 3.0, 4.0$, and 5.0 are coarse-grained n times; (b) $n = 1$, (c) $n = 2$, and (d) $n = 3$. Clearly shown is that the degree exponent does not change upon the iteration of coarse graining. Full lines in (a)-(c) are for the power-law distributions with the exponents 2.5 and 5.0 , while in (d) only the line for the exponent 2.5 is shown for comparison. γ values in (b)-(d) only indicate the degree exponents for the corresponding initial networks.

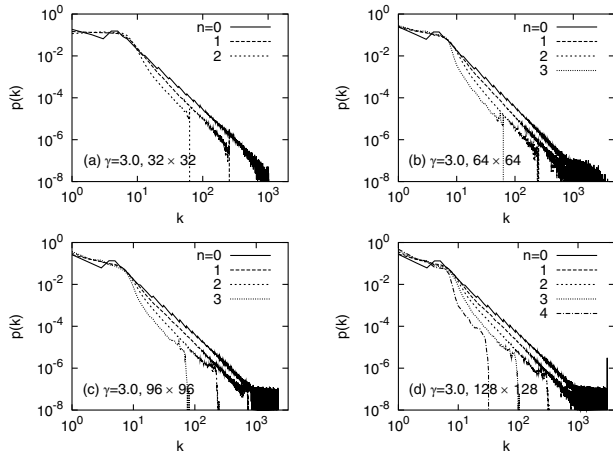


FIG. 3. Degree distribution $p(k)$ versus the degree k for the networks with the degree exponent $\gamma = 3$. The original network sizes are (a) 32×32 , (b) 64×64 , (c) 96×96 , and (d) 128×128 . The resulting coarse-grained network at the n th iteration displays the same degree exponent.

ing, considering the recent study in Ref. [5] that scale-free distributions can emerge from merging. Our main results [7] in the present study imply that this is not the case and that accordingly the scale-free distribution in human brain functional network is expected to be the genuine property of the brain, not the artifact of the coarse-grained information.

We next investigate other important structural properties of networks. Many real networks including Internet, World Wide Web, and the actor network, are characterized by the existence of hierarchical structure [8,9], which can usually be detected by the negative correlation between the clustering coefficient (see Ref. [2]) and the degree [8]. For example, the Barabási-Albert network [3], which does not possess hierarchical structure, is known to have the clustering coefficient C_v of the vertex v independent of its degree k_v [i.e., $C(k) \sim k^0$, see Ref. [8]], while the Holme-Kim model [10] has been shown to have $C(k) \sim k^{-1}$ [11], in accord with the observations of many real networks [8]. In Fig. 4, we plot $C(k)$ at the n th iteration step of the coarse graining for the initial network of the size 128×128 with $\gamma = 3$. The geographically embedded network in Ref. [6] is found to be somehow special since $C(k)$ is better described by $C(k) \sim k^{-2}$ rather than the abundantly found $C(k) \sim k^{-1}$. But this feature remains the same upon the coarse-graining procedure, implying that the coarse graining does not change the hierarchical structure of the network.

We next study the assortative mixing characteristics [12,13] of the network. For the assortative network, vertices with the higher degree tend to have high-degree neighbor vertices, while for the disassortative network, higher degree vertices favor lower degree neighbors. The degrees k_v and k_w of the two vertices v and w connecting each edge are measured and then the histogram is computed by using 20×20 bins in log-log scales in a k_v - k_w

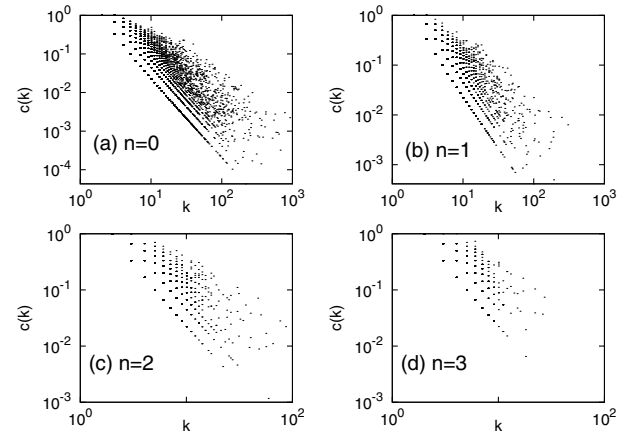


FIG. 4. The clustering coefficient $C(k)$ versus k for the 128×128 network with $\gamma = 3$ at the n th iteration step of coarse graining for (a) $n = 0$, (b) $n = 1$, (c) $n = 2$, and (d) $n = 3$. The qualitative feature remains the same upon coarse-graining procedure.

plane. The brightness of the region in Fig. 5(a)–5(c) is chosen in proportion to the logarithm of the height of the histogram in that region. Again we found that the coarse-graining procedure does not change the disassortative mixing property of the network; i.e., at any iteration step, the high-degree vertices in the network tend to have low-degree neighbors. This behavior of the disassortative mixing can also be detected by the assortativity coefficient r (see Ref. [12] for the definition). If r has positive value, the network has assortative mixing property while it is disassortative otherwise. In Fig. 5(d), r is shown to have negative values at the $n = 0, 1, 2$, coarse-

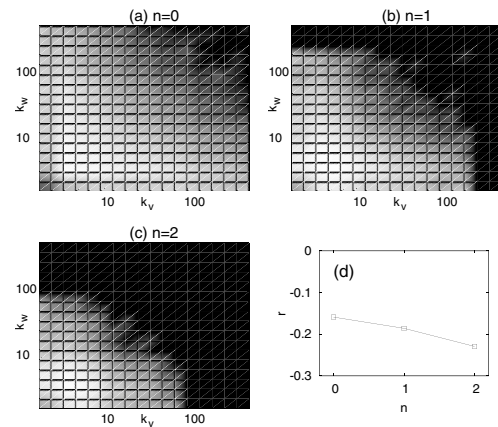


FIG. 5. (a)-(c) Density plot of the degrees k_v and k_w where v and w are two vertices connected by each edge. A brighter region indicates that there are more edges in that region. The initial network shows disassortative behavior (higher k_v prefers lower k_w , and vice versa), which remains qualitatively the same as the coarse-graining procedure proceeds n times: (a) $n = 0$ (initial network), (b) $n = 1$, and (c) $n = 2$. (d) Assortativity coefficient r versus the number n of iterations of coarse graining. After the coarse graining, the network remains to be disassortative.

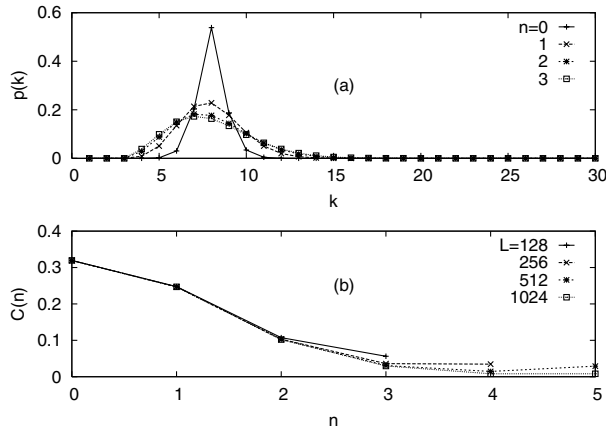


FIG. 6. Geographically embedded WS network at the rewiring probability $P = 0.1$. (a) Degree distribution $p(k)$ at the iteration steps $n = 0, 1, 2$, and 3 . (b) Clustering coefficient $C(n)$ versus n .

graining steps. The decrease of r with n is not completely understood, although this dependence of r versus the network size N appears to be consistent with Ref. [12], where r tends to approach zero from below as the larger disassortative network is considered.

So far we have introduced a geographical coarse-graining procedure and applied it to the geographically embedded scale-free network in Ref. [6]. Although the network sizes become smaller as the coarse-graining procedure proceeds, it has been found that several key features of the initial networks do not change qualitatively. In particular, the degree exponent γ does not change, and the hierarchical structure [detected by the negative correlation between the clustering coefficient $C(k)$ versus degree k] and the disassortativity (detected by more edges connecting high-degree vertices to low-degree vertices than to high-degree vertices) remain qualitatively the same. Our geographic coarse-graining procedure can be useful when the initial network is of a huge size since one can then systematically reduce the network size without destroying important characteristics of the network. Modification of the present coarse-graining method to apply for the network which is not geographical embedded can be an interesting extension. The main results also suggest that the scale-free distribution found recently for the human brain functional network may not be an artifact due to the large voxel size (each contains $O(10^5)$ neuron cells), but the genuine property of the brain.

We finally study the two-dimensional Watts-Strogatz (WS) network [14], built similarly to Ref. [2]: Vertices are put on the two-dimensional square lattice points and every vertex is connected to its nearest and next-nearest neighbor vertices. Each edge is visited once, and with the rewiring probability P is rewired to a randomly chosen vertex. The resulting network belongs to the so-called exponential network since the tail in the degree distribu-

tion is exponentially small. We then iterate our geographic coarse-graining procedure with the average degree kept constant at each iteration. In Fig. 6(a), the initial network of the size 128×128 at the rewiring probability $P = 0.1$ is coarse-grained n times. As n becomes larger, the degree distribution remains exponential and tends to saturate. In Fig. 6(b), initial two-dimensional WS networks of various sizes $L = 128, 256, 512$, and 1024 at $P = 0.1$ are coarse-grained and the clustering coefficient $C(n)$ is plotted as a function of the number n of iterations. As the coarse-graining process proceeds the clustering coefficient is shown to decrease towards zero, which indicates that the RG stable fixed point of the WS network is close to the random network of Erdős and Rényi.

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- [1] For general reviews, see, e.g., D. J. Watts, *Small Worlds* (Princeton University Press, Princeton, 1999); R. Albert and A.-L. Barabási, *Rev. Mod. Phys.* **74**, 47 (2002); S. N. Dorogovtsev and J. F. F. Mendes, *Evolution of Networks* (Oxford University Press, New York, 2003).
 - [2] D. J. Watts and S. H. Strogatz, *Nature (London)* **393**, 440 (1998).
 - [3] A.-L. Barabási and R. Albert, *Science* **286**, 509 (1999); A.-L. Barabási, R. Albert, and H. Jeong, *Physica A (Amsterdam)* **272**, 173 (1999).
 - [4] V. M. Eguíluz, D. R. Chialvo, G. Cecchi, M. Baliki, and A. V. Apkarian, *cond-mat/0309092*.
 - [5] B. J. Kim, A. Trusina, P. Minnhagen, and K. Sneppen, *nlin.AO/0403006*.
 - [6] A. F. Rozenfeld, R. Cohen, D. ben-Avraham, and S. Havlin, *Phys. Rev. Lett.* **89**, 218701 (2002); D. ben-Avraham, A. F. Rozenfeld, R. Cohen, and S. Havlin, *Physica A (Amsterdam)* **330**, 107 (2003).
 - [7] The main results in this work are obtained for the networks embedded in two dimensions while the human brain network is embedded in three dimension. The extension of the present study to a higher dimension can be interesting.
 - [8] E. Ravasz and A.-L. Barabási, *Phys. Rev. E* **67**, 026112 (2003).
 - [9] A. Trusina, S. Maslov, P. Minnhagen, and K. Sneppen, *Phys. Rev. Lett.* **92**, 178702 (2004).
 - [10] P. Holme and B. J. Kim, *Phys. Rev. E* **65**, 026107 (2002).
 - [11] G. Szabó, M. Alava, and J. Kertész, *Phys. Rev. E* **67**, 056102 (2003).
 - [12] M. E. J. Newman, *Phys. Rev. Lett.* **89**, 208701 (2002); M. E. J. Newman and J. Park, *Phys. Rev. E* **68**, 036122 (2003).
 - [13] S. Maslov and K. Sneppen, *Science* **296**, 910 (2002).
 - [14] Similar study has also been made in M. A. de Menezes, C. F. Moukarzel, and T. J. P. Penna, *Europhys. Lett.* **50**, 574 (2000).